



KlimaGuard Kids

KGK Technical Commendation

Sustainow Technologies · MIT Open Source

Rich-text technical commendation for partners & procurement

Field	Detail
Document title	KlimaGuard Kids — KGK Technical Commendation
Product name	KlimaGuard Kids
Supplier / developer	Sustainow Technologies
Document version	1.0
Software version	0.2.0
License	MIT (open source)
Primary repository	https://github.com/carbonintelli/KlimaGuardKids
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Classification (indicative UNSPSC)	43230000 Software; 81112200 Software maintenance and support; 77101500 Environmental management; 85101705 Public health administration

Purpose of this document This document provides a clear technical description of KlimaGuard Kids (KGK) for procurement, ICT, programme, partner, and evaluation reviewers. It serves as the product technical commendation and capability statement.

1. Executive summary

KlimaGuard Kids is an open-source, web-based platform that converts live climate and air-quality signals into **child-centred health preparedness guidance**. It uses eight cooperating software “agents” (deterministic TypeScript modules) to analyse heat, air quality, flood/drought pressure, nutrition stress, and disease risk, then produces:

- 1 Location-specific climate–health briefings for caregivers, schools, and community health workers
- 2 Age-banded guidance for children (ages 5–8, 9–12, 13–17)
- 3 For India, a transparent **Child Health Impact Score (CHIS)** across five measurable dimensions
- 4 An optional age-appropriate **play / mission** layer to reinforce preparedness behaviours

The platform is designed for **anticipatory action**, not clinical diagnosis. All core measurement logic is inspectable in source code. Live weather and air-quality inputs are drawn from Open-Meteo; health framing references WHO and national public-health guidance.

Coverage (demo registry):

Scope	Count
Climate-vulnerable countries	159
High-risk city presets (Tier 1–3)	482
India regional zones (deep coverage)	77
Validated / trusted data sources	50

2. Problem statement and development relevance

Climate change intensifies heatwaves, flooding, air pollution, vector-borne disease, and food insecurity. Children are disproportionately affected because of physiology, dependency on caregivers, and disruption to school and nutrition routines. Most climate tools speak to policymakers; few deliver **actionable, age-appropriate guidance** that local actors can use within days.

KlimaGuard Kids addresses this gap by:

- Connecting **near-real-time climate data** to child-health heuristics
- Presenting outputs in language suitable for children and supervising adults
- Providing a **quantified India impact score** for regional prioritisation
- Remaining **open source (MIT)** so UN entities, governments, NGOs, and local partners can adapt, host, and extend it

2.1 Sustainable Development Goal alignment

SDG	Contribution
SDG 3 — Good health and well-being	Anticipatory guidance for climate-sensitive child health risks
SDG 13 — Climate action	Child-centred early-warning and preparedness intelligence
SDG 2 — Zero hunger	Nutrition and food-security stress signals under climate disruption
SDG 4 — Quality education	School-ready heat/flood/air guidance and peer-educator missions for teens
SDG 11 — Sustainable cities	City- and region-level vulnerability views for local planning

3. Solution overview

3.1 What the product delivers

Capability	Description
Global dashboard	Select a vulnerable country and city; run the full agent pipeline; view risks, correlations, and child guidance
India dashboard	Select one of 77 regions; view regional monsoon/zone context and CHIS composite + dimension scores
Kids play mode	Age-tiered missions, badges, levels, and streaks derived from preparedness tips (browser-local progress)
Machine-readable API	REST endpoints for countries, India regions, and full analysis reports (JSON)
Provenance	Each agent run cites trusted data/reference sources

3.2 What the product is not

- Not a medical diagnostic or telemedicine system
- Not a substitute for clinical care, emergency services, or national early-warning authorities
- Not a system that requires child accounts or stores children’s personal health records in the demo architecture

3.3 Intended users

User group	Typical use
Community health workers / frontline staff	Quick location briefings before heat/flood/vector seasons
Teachers and school administrators	Age-appropriate preparedness messaging
Caregivers and youth (supervised)	Play missions and simple guidance cards
Programme / UN / NGO officers	Situational awareness, demo for pilots, adaptation of open code
Local technologists	Extend registries, deploy in-country, integrate APIs

4. Technical architecture

Detailed diagrams are provided in [docs/ARCHITECTURE.md](#) and [docs/images/](#).

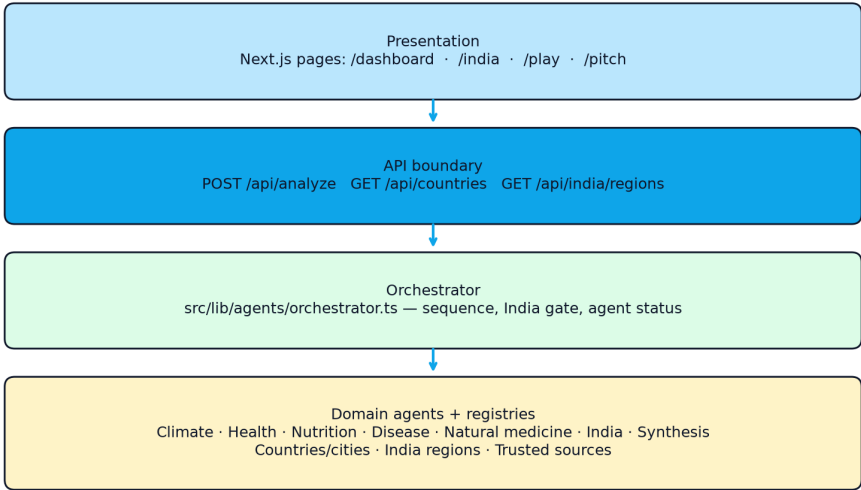
4.1 Architectural style

KlimaGuard Kids is a **modern web application** with:

- **Presentation layer** — Next.js App Router pages (React)
- **API layer** — Next.js Route Handlers (`/api/*`) with Zod validation
- **Domain layer** — Orchestrated agent pipeline in TypeScript
- **Data layer** — Static geographic registries + live Open-Meteo climate/AQ fetch

Browser UI → HTTPS JSON APIs → Orchestrator → Agents → SynthesisReport → UI

System layers — UI → API → Orchestrator → Agents → Report



Output contract: SynthesisReport JSON → ReportView / IndiaImpactPanel / KidsPlayHub

System layers diagram

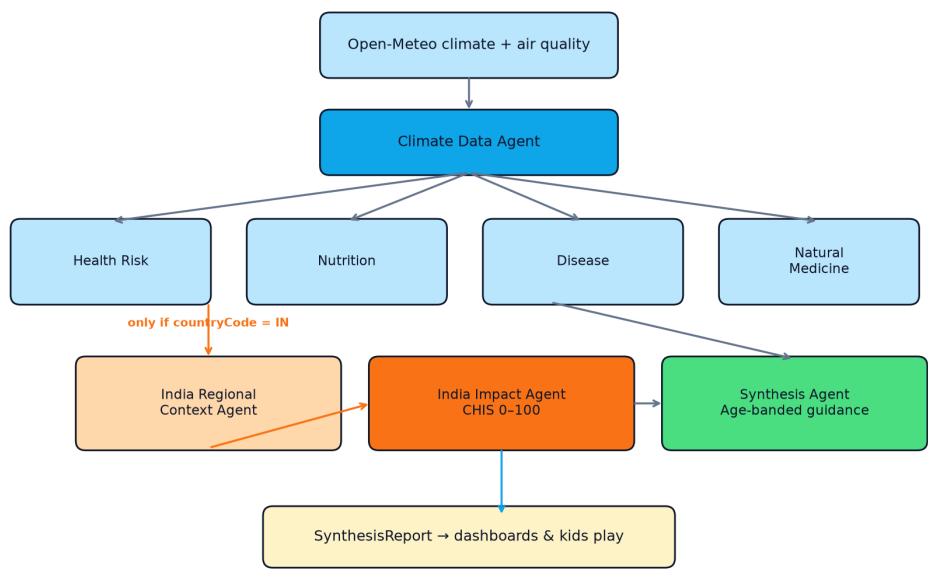
4.2 Technology stack

Layer	Technology	Rationale
Application framework	Next.js 15 (App Router)	Unified UI + API, strong TypeScript support

Layer	Technology	Rationale
UI	React 19, Tailwind CSS 4	Accessible, responsive interfaces
Language	TypeScript (strict)	Type-safe contracts for reports and APIs
Validation	Zod	Explicit request schemas for /api/analyze
Live climate data	Open-Meteo Forecast & Air Quality APIs	Public, no API key required for demo scale
Quality gates	ESLint + next build via GitHub Actions CI	Automated lint/build on every push/PR
License	MIT	Permissive reuse by UN system and partners

4.3 Agent pipeline (eight agents)

Eight-agent pipeline — global path vs India path



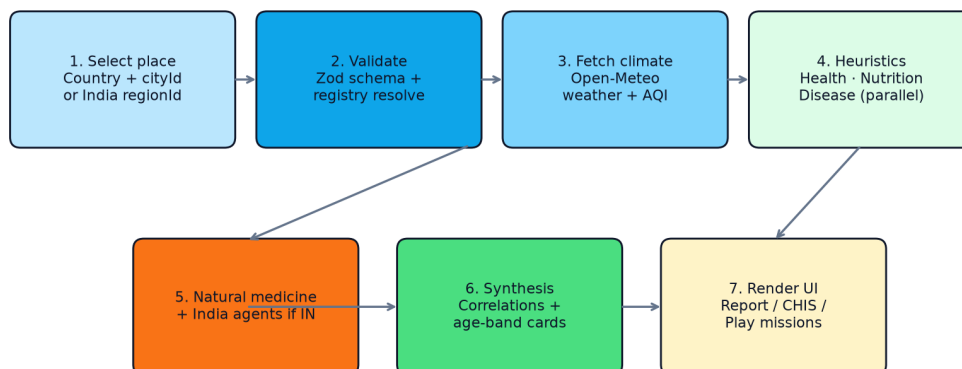
Eight-agent pipeline

#	Agent	Scope	Function
1	Climate Data Agent	Global	Fetches temperature, humidity, precipitation, heat index, AQI
2	Health Risk Agent	Global	Maps climate signals to heat, respiratory, flood, vector stressors
3	Nutrition & Food Security Agent	Global	Hydration targets, food safety, scarcity notes
4	Disease Outlook Agent	Global	Transmission pathways, precautions, illness profiles
5	Natural Medicine Agent	Global	Evidence-tagged supportive home remedies under adult supervision
6	India Regional Context Agent	India only	Monsoon, climate zone, regional child vulnerability
7	India Child Health Impact Agent	India only	Computes CHIS (0–100) across five dimensions
8	Synthesis Agent	Global	Cross-agent correlations + age-banded child guidance cards

Important design choice: In the current demo pipeline, agents are **deterministic TypeScript modules** with documented heuristics—not opaque large-language-model calls. This supports auditability, reproducibility, and offline/local adaptation.

4.4 Request processing flow

Analyze sequence — from city selection to child guidance



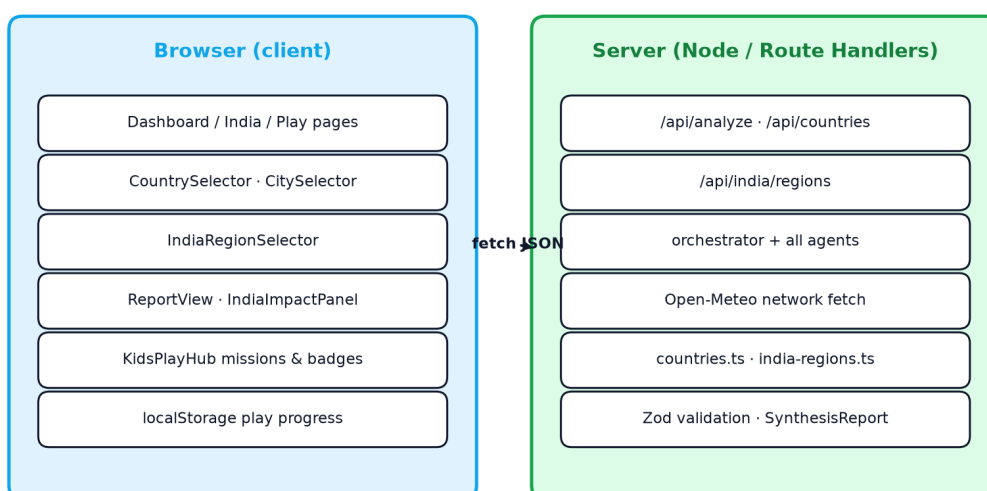
Failure mode: climate fetch error → HTTP 500. Missing country/city → HTTP 404.

Analyze sequence diagram

- 1 User selects a country + city (or an India region).
- 2 Client calls `POST /api/analyze` with `{ countryCode, cityId? }` or `{ countryCode: "IN", regionId }`.
- 3 Server validates input (Zod) and resolves latitude/longitude from registries.
- 4 Orchestrator fetches climate data; on failure returns HTTP 500.
- 5 Health, nutrition, and disease agents run in parallel; natural-medicine follows.
- 6 If India: regional context → CHIS impact scoring.
- 7 Synthesis assembles `SynthesisReport` JSON.
- 8 UI renders dashboards; play mode may convert guidance into missions.

4.5 Client–server boundary

Client vs server boundary



Client versus server responsibilities

Server-side	Client-side
All agent computation	Dashboards, selectors, report rendering
Open-Meteo network fetch	Kids play interactions
Registry resolution for analysis	Play XP/badges/streaks in localStorage
API validation and error responses	No child account system in demo

5. Functional modules and user interfaces

Route	Module	Description
/	Landing	Product introduction and entry points
/dashboard	Global analysis	Country + vulnerable-city selection; full report
/india	India CHIS	77-region selector; impact panel and guidance
/play	Kids play	Age bands 5–8 / 9–12 / 13–17; missions and badges
/pitch	Stakeholder brief	Narrative overview for partners

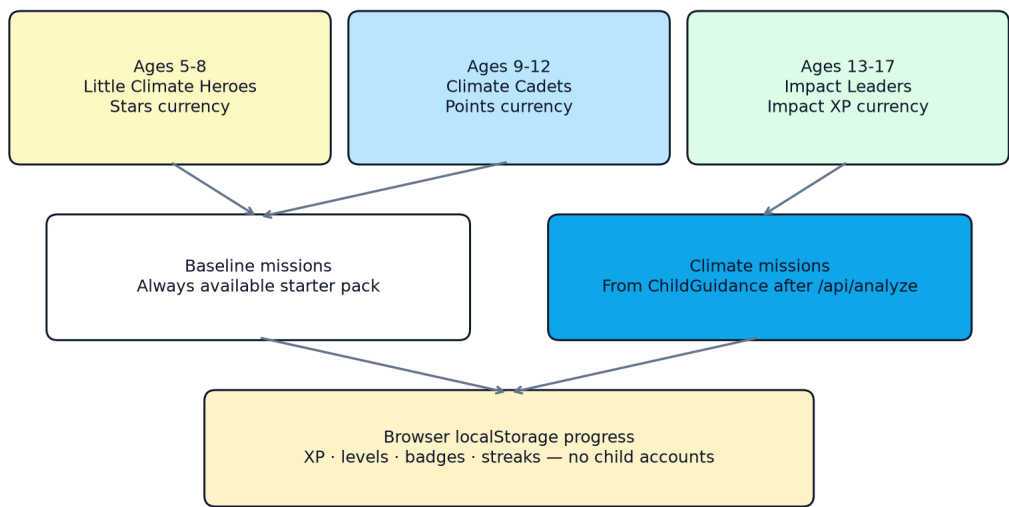
5.1 Age-banded child guidance

The Synthesis Agent produces three guidance cards per analysis:

Age band	Tone	Example focus
5–8	Playful, simple	Shade, water sips, handwashing
9–12	Practical briefing	Forecast checks, cooler routes, peer reminders
13–17	Leadership / community	Buddy checks, verified alert sharing, peer education

5.2 Kids play (behaviour reinforcement)

Kids play — age-tiered missions from guidance



Kids play gamification model

Ages	Mode name	Currency	Progress storage
5–8	Little Climate Heroes	Stars	Browser only
9–12	Climate Cadets	Points	Browser only
13–17	Impact Leaders	Impact XP	Browser only

Missions can use a starter pack or unlock from live `ChildGuidance` after climate analysis. **No child accounts** are required for the demo.

6. Data sources, geography, and methodology

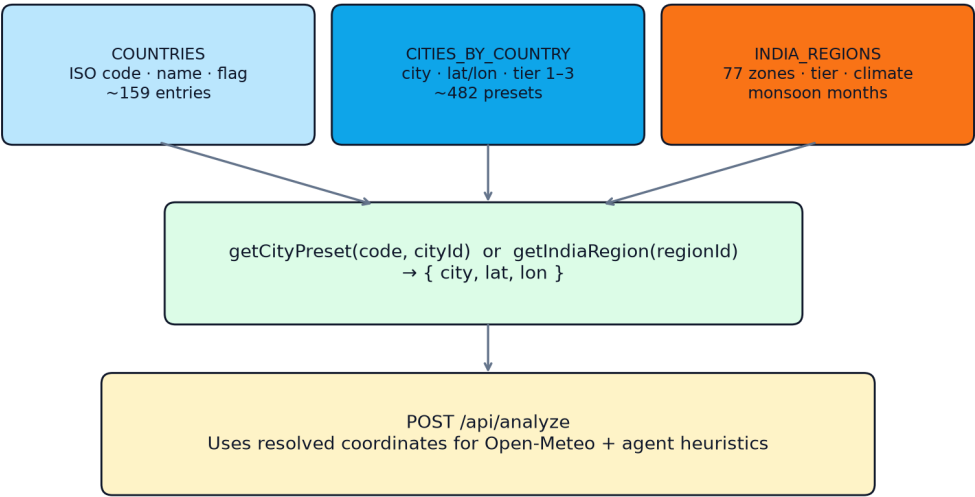
6.1 Trusted sources

Live operational fetch today uses Open-Meteo. Remaining entries are authoritative reference frameworks that ground agent heuristics, India CHIS notes, and UN-aligned provenance (see `src/lib/sources.ts` for the full catalogue of ~50 sources).

Source family	Examples	Use in platform
Live weather / AQ	Open-Meteo Forecast & Air Quality	Near-real-time temperature, precip, humidity, AQI
Climate EO / forecast	Copernicus CDS, NASA POWER, CHIRPS, ECMWF, NOAA CPC, ClimateSERV	Climate context and early-warning framing
UN health & child rights	WHO heat-health / WASH / GHO, UNICEF CCRI / WASH / MICS	Child-health and WASH heuristics
Disaster & adaptation	UNDRR Sendai, INFORM, ND-GAIN, EM-DAT, ReliefWeb	Risk and vulnerability framing
Food security	FAO / GIEWS, WFP, IPC, FEWS NET	Nutrition and scarcity stress framing
Water / development	WRI Aqueduct, World Bank Climate Portal	Water-stress context
Regional health / met	PAHO, Africa CDC, ECDC, CIMH, SPREP, Met Office, BoM, JMA, US CDC	Region-specific disease and climate services
Humanitarian context	ACAPS, UNHCR climate, IOM DTM	Displacement and crisis vulnerability
India national	IMD, CPCB, NFHS-5, NVBDCP, IDSP, NCDC	India regional and CHIS scoring context

6.2 Geographic registries

Data registries — how places resolve to coordinates



Data registries diagram

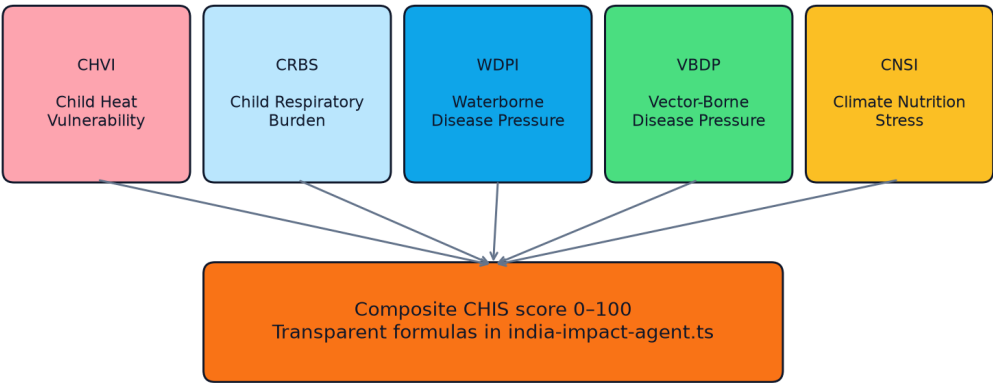
Registry	Location in code	Contents
Countries	src/lib/countries.ts	159 ISO countries with flags
Cities	CITIES_BY_COUNTRY in same file	482 city presets with lat/lon, urban tier (1–3), and primary risk tags
India regions	src/lib/india-regions.ts	77 zones with tier, climate zone, monsoon months, primary risks
Trusted sources	src/lib/sources.ts	~50 validated feeds and reference frameworks

City presets emphasise climate-vulnerable locations (heat, flood, cyclone/typhoon, drought, air pollution, sea-level, vector pressure) across metro (Tier 1), emerging (Tier 2), and district/frontier (Tier 3) centres. India retains a dedicated regional model for deeper CHIS analysis.

For the maintained catalogue (tier definitions, helper functions, source families, and orchestrator provenance mapping), see [DATA SOURCES AND GEOGRAPHY.md](#).

6.3 India Child Health Impact Score (CHIS)

India CHIS — five child-health impact dimensions



Activated only for countryCode = IN after regional context agent runs

CHIS dimensions diagram

CHIS is a **0–100 composite** (higher = greater climate-related child-health burden) with five dimensions:

Code	Dimension	Focus
CHVI	Child Heat Vulnerability Index	Extreme heat days, heat index, aridity
CRBS	Child Respiratory Burden Score	AQI and air-pollution pressure
WDPI	Waterborne Disease Pressure Index	Flood / water-contamination pressure
VBDP	Vector-Borne Disease Pressure	Conditions favouring vector risk
CNSI	Climate Nutrition Stress Index	Climate-linked nutrition/food stress

Formulas are transparent and open source in `src/lib/agents/india-impact-agent.ts`. Risk bands used in UI: low / moderate / high / critical.

7. Application programming interfaces (APIs)

Base URL (local demo): `http://localhost:3000`

Endpoint	Method	Description
<code>/api/countries</code>	GET	Countries with nested city presets and registry counts
<code>/api/india/regions</code>	GET	77 Indian regions with metadata
<code>/api/analyze</code>	POST	Runs full agent pipeline; returns <code>SynthesisReport</code>
<code>/api/analyze</code>	GET	Service metadata (agents, country/city/region counts)

7.1 Example — analyse a vulnerable city

```
curl -X POST http://localhost:3000/api/analyze \
  -H "Content-Type: application/json" \
  -d '{"countryCode": "BD", "cityId": "chattogram"}'
```

7.2 Example — analyse an India region

```
curl -X POST http://localhost:3000/api/analyze \
  -H "Content-Type: application/json" \
  -d '{"countryCode": "IN", "regionId": "mumbai"}'
```

7.3 Interoperability notes

- Responses are **JSON** suitable for integration into dashboards, SMS gateways (via partner middleware), or national platforms.
- Optional overrides: `city`, `lat`, `lon` may be supplied to analyse coordinates beyond the demo registry.
- No proprietary binary protocols; standard HTTPS/JSON only.

8. Security, privacy, and child safeguarding

Topic	Current demo posture
Authentication	Public demo; no end-user login required
Child accounts	Not used — play progress is device-local (<code>localStorage</code>)
Personal data	Analysis uses country/city/region only; no name, phone, school, or health-record collection in core flow

Topic	Current demo posture
Server-side storage of chat/PII	Not part of current architecture (chat/scheduling removed)
Transport	Deploy behind HTTPS in production
Secrets	Open-Meteo demo path requires no API key; production integrations should use environment-managed credentials if added
Safeguarding principle	Outputs are educational/preparedness guidance; clinical decisions require qualified providers
Future compliance path	COPPA / GDPR-K / national child-data laws for any future account or messaging features

Disclaimer for procurement reviewers: KlimaGuard Kids is a **community preparedness and decision-support tool**, not a medical device and not a diagnostic system.

9. Deployment, hosting, and operational requirements

9.1 Minimum runtime

Requirement	Specification
Runtime	Node.js 22 (CI baseline)
Package manager	npm
Build	npm ci && npm run build
Start	npm start (or npm run dev for development)
Outbound network	Access to Open-Meteo endpoints for live analysis

9.2 Typical deployment options

- Cloud VM / container (Dockerisable Next.js app)
- Managed Node hosting (e.g. Vercel, equivalent UN/partner cloud)
- On-premises / in-country government cloud for data-sovereignty pilots

9.3 Quality assurance

Automated CI (GitHub Actions) on `main` and pull requests:

```

1  npm ci
2  npm run lint
3  npm run build

```

9.4 Extensibility for UN programmes

Partners can:

- Add countries/cities in `src/lib/countries.ts`
- Add India (or analogous national) regions in dedicated registries
- Register new agents in `src/lib/agents/` + orchestrator
- Re-skin UI for local languages (i18n roadmap)
- Replace Open-Meteo with authenticated national meteorological feeds

See [CONTRIBUTING.md](#).

10. Intellectual property, licensing, and openness

Item	Detail
Copyright	© 2026 Sustainow Technologies
License	MIT License — use, modification, redistribution permitted
Source availability	Full application source on GitHub
Third-party data	Open-Meteo and public reference sources retain their own terms; platform cites provenance
Brand assets	Product logo and documentation diagrams included in repository

Open licensing is intentional: UN entities and implementing partners can fork, audit, localise, and redeploy without proprietary lock-in for the core demo codebase.

11. Implementation support model (indicative)

For UNGM capability statements, the following support tiers are typical (commercial terms negotiated per engagement):

Tier	Scope
Open-source self-serve	Public repo, documentation, community issues
Pilot support	Deployment assistance, registry customisation, training for programme staff
Country adaptation	Local language UI, national met/health feed integration, hosting hardening
Scale / LTA-ready	SLA-backed hosting advice, monitoring, change management, knowledge transfer

Training assets are also available under *Training/* (marketing/training manual and diagrams).

12. Risks, limitations, and roadmap

12.1 Current limitations

- Heuristic health models are **decision-support**, not epidemiological forecasting validated for clinical use
- Global city registry is a curated demo set (482 cities across Tier 1–3), not exhaustive
- India has deeper quantitative scoring than other countries today
- Multilingual UI and offline-first modes are roadmap items
- Production authentication, RBAC, and enterprise SSO are not in the public demo

12.2 Indicative roadmap

- 1 Local language packs and accessibility hardening
- 2 Optional authenticated national weather/AQ feeds
- 3 Expanded non-India quantitative impact modules
- 4 Partner SMS/USSD or WhatsApp bridges via middleware (privacy-reviewed)
- 5 Field evaluation with ministries of health/education and UN programme partners

13. Document map for reviewers

Document	Contents
This file (docs/KGK_Technical_Commendation.pdf)	KGK technical commendation / capability statement
docs/ARCHITECTURE.md	Concise design/architecture with diagrams
docs/DATA_SOURCES_AND_GEOGRAPHY.md	Tiers, registry counts, validated sources, agent provenance
docs/images/	Technical diagrams (system layers, pipeline, CHIS, etc.)
README.md	Product overview and quick start
CONTRIBUTING.md	How to extend geography, sources, and agents
LICENSE	MIT licence text
Training/	Longer narrative and presentation assets

13.1 Diagram index

File	Explains
docs/images/system-layers.png	UI → API → orchestrator → agents
docs/images/analyze-sequence.png	End-to-end analyse request steps
docs/images/agent-pipeline.png	Global vs India agent flow
docs/images/client-server.png	Browser vs server responsibilities
docs/images/data-registries.png	Country/city/India resolution
docs/images/chis-dimensions.png	CHIS five dimensions
docs/images/play-gamification.png	Age-tiered play model
docs/images/repo-map.png	Code ownership map

Regenerate diagrams:

```
python3 docs/generate_architecture_diagrams.py
```

14. Contact and submission notes

Item	Detail
Supplier	Sustainow Technologies
Product	KlimaGuard Kids
Repository	https://github.com/carbonintelli/KlimaGuardKids
Licence	MIT

Distribution guidance: Use this PDF (docs/KGK_Technical_Commendation.pdf) as the official KlimaGuard Kids technical commendation for partner, procurement, and programme submissions. Diagrams in docs/images/ are embedded below where relevant.

End of KGK Technical Commendation.